

# *An Approach Towards Building an IoT Based Smart Classroom*

Ani R, Krishna S, Akhil H, Arun U

Department of Computer Science & Applications

Amrita Vishwa Vidyapeetham, Amritapuri, India

[anir@am.amrita.edu](mailto:anir@am.amrita.edu), [krishnas@am.amrita.edu](mailto:krishnas@am.amrita.edu),

[akhilkmr18@gmail.com](mailto:akhilkmr18@gmail.com), [arunudayan2013@gmail.com](mailto:arunudayan2013@gmail.com)

**Abstract**—IoT (Internet of Things) is a dynamic innovation with a powerful impact on today's world which can make human life simple and effortless. The scope of this field is limitless and has emerged as a winner in various areas ranging from Medicine, Engineering, Computer Science, Space and Technology, Automobiles and so on. The center of purpose is utilizing IoT based technology in accomplishing automation for classrooms. In this paper, we propose an approach to control and manage electrical equipments such as fans and lights based on human presence. Our focus is towards building a solution which could help in reducing overutilization of energy resources. A camera is used for recognizing the presence of people in the classroom and for analyzing their seating position. Here a classroom is divided into two segments. Whenever a human presence is detected in a particular segment then the light and fan will be switched ON. The reasonable objective of this paper is how to build up a smart classroom where we can automate the electrical equipments with a focus towards energy conservation.

**Keywords**—IoT, Camera-Based Detection, Smart classroom, Automation.

## I. INTRODUCTION

The Internet of Things (IoT) is a technology solution which has the ability to create revolution in upcoming years. This technological revolution could help us in redefining our living standards by providing automation, one of the most commonly used terms in the field of science and technology [1]. Automation could help us in achieving complete control over the electric equipments surrounding us at our home or work environment. The day is not far away when all our equipments would have the ability to communicate with us. We could control them either through voice based commands or through gestures based techniques [2]. IoT will help in building an intelligent system that could understand our requirements and function accordingly without even the need to command them. In this paper the main objective centers on implementing IoT technology for building a smart classroom environment. Our focus is towards finding the issues

confronted and proposing solutions for the same. We choose classroom as an environment because it is a place where we could easily identify the problems. We saw students and teachers switch ON the lights and fans but most of the time they forget to switch it OFF. With our work we try to overcome such situations by finding solutions where our environment can be controlled and automated.

Recent advances in computer systems have created the necessary conditions for a wide variety of devices. The initial idea was using sensors such as Passive Infrared Sensor (PIR) and touch sensors to detect human presence and accordingly control the classroom environment. We built a simulated version of the actual environment to understand the working. But it had certain disadvantages such as availability of required sensor, cost, regular wear and tear and user discomfort. So, we came up with the idea of using a camera based detection which is cheaper and easier to implement.

## II. RELATED WORK

Nowadays IoT based automation has a great impact on many sectors. Many of them implement this technique for managing electrical equipments and marking attendance [3]. Classrooms are the place which could be more suitable to implement this technique because it's a place where more energy wastage occurs [4]. Here a system was implemented to conserve energy using IoT based automation technique and they used different sensors such as LDR and LM35 for detecting the human presence [5]. When studying about different sensors we realized that sensors have many demerits such as - lifetime of sensors, cost, connection issues and the results produced cannot always be reliable. We thought about different ways for overcoming the limitations with sensors and decided to use camera for detecting human presence. There are many attendance marking systems based on face detection techniques using camera [6]. Comparatively camera will give better results than sensors because cameras are build to detect human presence more accurately whereas sensors need to be customized according to different user requirements and their use may vary from weather forecasting, disease prediction [7]

and healthcare systems [8]. Using this IoT enabled technique we can also avoid energy wastage [9].

In this paper we describe how to control and manage electrical equipments using camera based human detection techniques along with attendance marking system.

## II. METHODOLOGY

Inside a classroom the electric equipments, for example, fans and lights are controlled utilizing camera-based discovery procedures. A solitary camera is used for person detection. We partition the image frame into two different sub frames. These sub frames are to depict different areas of the classroom with each segment containing tables and chairs arranged in a vertical manner with fan and light. The human presence is captured using a camera and the image is processed with OpenCV and python. When the human presence is detected for a frame segment then light and fan assigned for that frame will be automated or else they remain in OFF state. By using automation technique we can maintain a strategic distance from energy wastage in our classrooms which occurs mostly when we forget to switch OFF the electrical equipments after their usage.

A proposed classroom environment is depicted with person surveillance done using a single camera (Fig.1).



Fig 1. A Proposed Classroom Environment

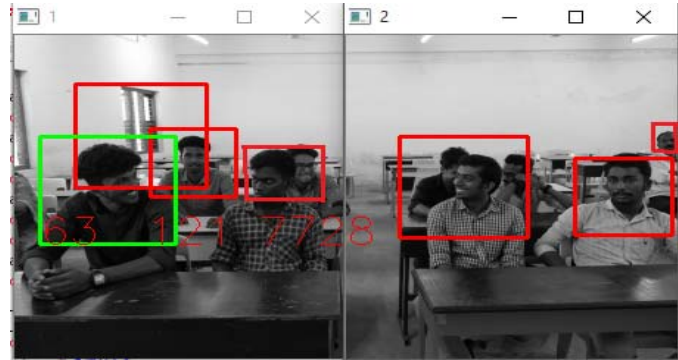


Fig 2. Proposed classroom environment 2 Frame Architecture

Fig.2 illustrates a proposed classroom environment. We divided the classroom into two segments. Students are seated accordingly. We are detecting presence using haar cascade classifier which is an object detection algorithm based on 3 feature extraction techniques including modules such as edge features, line features ,four rectangle feature[10]. This cascade classifier can be used to detect features such as facial landmarks, board signs, pedestrian detection and so on which can be applied in an image or video. Haar cascades have certain advantages such as maximum accuracy in face detection also the false positive rate in object detection is low. In Fig.2 detected faces are marked inside a rectangular box using cascade classifier.

Fig.3 shows the classroom architecture where the entire classroom is monitored using a single camera. If human presence is detected in frame 1 then the electric equipments assigned for that area will start working automatically .The same follows for the other areas. Once students leave the classroom all the electric equipments will be automatically switched OFF.

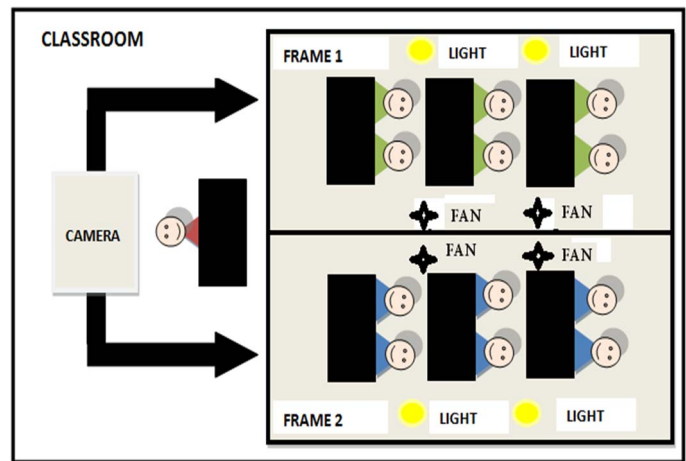


Fig 3. Arranged Classroom Environment Architecture

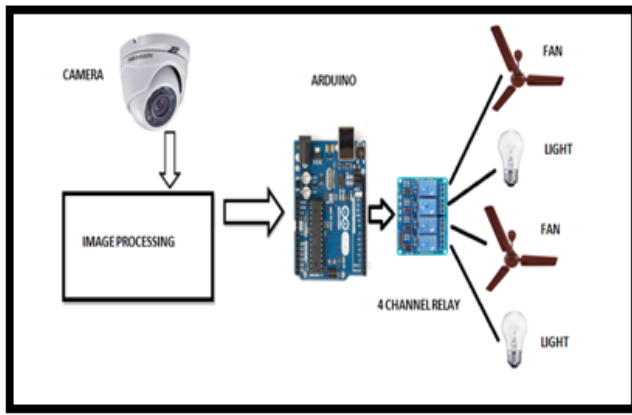


Fig 4. Architecture diagram for Arduino control

#### PSEUDO CODE

```

-Initialize the face value to 0
-If presence is detected in a position
set face value to 1

If(len(faces))==1:

-Set the serial value to the assigned
variable
{
Myserial.write('Y')
}
-If human is not detected set face value
to 0

If(len(faces))==0:

-Set the serial value to the assigned
variable
{
Myserial.write('N')
}
-If value=0 Set PinMode = LOW
If(Serial.read()=='N')
{
digitalWrite(PinNo,PinMode)
-Set the PinMode to OFF state If
the value read is 0
setPinMode=LOW
}
-If value=1 Set PinMode = HIGH
If(Serial.read()=='Y')
{
digitalWrite(PinNo,PinMode)
set the PinMode to ON state If the
value read is 0
setPinMode=HIGH
}
  
```

#### IV. PROPOSED HARDWARE SETUP

The system framework structure incorporates Arduino Uno microcontroller, 4 channel relay, power switch and lamp. Arduino is connected to PC. We are using PC camera for demonstration purposes. Camera detects human presence in a room. When presence is detected then the lights and fans for that room which are controlled using Arduino will be switched ON.

Similarly, when person leaves the room the lights will be switched OFF. Lamp is connected to Arduino using a relay module. This Relay channel controls the working of the electric equipments. Also, we are using Relay to connect with high voltage power source as the voltage range permitted for Arduino is 3v to 5v. Arduino cannot be directly connected with high voltage devices.

Fig.5 shows the hardware setup in which we use Arduino Uno ATMEGA328 and electrical equipments are controlled using the Arduino board through a 4 channel relay module. The relay will act as switch.

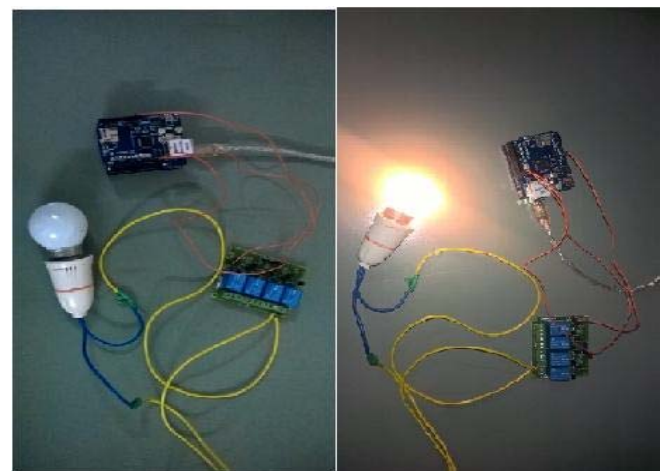


Fig 5. Proposed Hardware setup (Active & Dormant state)

Following components were used for creating the proposed project environment:

#### A. ARDUINO



Fig 6. Arduino Board



Arduino (*Fig.6*) is an open-source hardware & software platform which provides a straightforward method to control electric equipments. It's a small PC framework system that can be programmed with instructions to communicate with different forms of input and output. This controller has numerous simple and advanced IO pins with a power supply ranging from 5V to 20V, that can be powered either using a USB port or an external power source. It includes an ATmega328 microcontroller having many features such as 32KB of flash memory, 2 KB SRAM and 1 KB EEPROM.

### B. CAMERA



Fig 7. Camera

Camera (*Fig.7*) is a significant component because it captures videos and images. They resemble ordinary cameras but with a few additional features including monitors and wireless transmission facilities. These cameras are mainly utilized for surveillance purposes with a noteworthy usage at working environments and educational institutions. The camera will be dome camera. The main advantage is the image clarity and night vision facilities which make the recordings and pictures captured more precise and clear.

### C. RELAY



Fig 8. Relay Board

Relay (*Fig.8*) is an electrically operated device which acts as a switch for automatically controlling electrical equipments. Relay provides advantages such as stability, long-duration and reliability which are widely used device for power protection, automation technology and remote control and power electronics. Relay contains an indicator light which gives the status of the relay board (ON/OFF). The maximum output

voltage varies around DC 30V/10A, AC 250V/10A. The size and weight of the relay board is around 75mm (length) x 55mm (breadth) x 19.3mm (height). The life expectancy of a general purpose relay is rated to be 1,00,000 operations on minimum. The mechanical life expectancy may be a million, 10 million or even 100 million operations.

## V. RESULTS AND ANALYSIS

We are predicting an output using single camera based system. For demonstration purposes we used PC camera which can be upgraded to CCTV camera for better surveillance.



Fig 9. Hardware setup (in active state when presence is detected)

*Fig.9* shows the system in active state. Whenever a human presence is detected by the camera the lamp is automatically switched ON. Similarly if no human presence is detected by the camera the lamp is automatically switched OFF.

## VI. CONCLUSION

This project proposes a low cost, universally accessible, auto-configurable solution. The method discussed in this paper is original and has achieved the ability to control electrical equipments remotely. Camera based person detection provides a capable solution and it is low cost as compared to sensors. The proposed system provides scalability and flexibility. It can be effortlessly executed in all schools, universities and work environments.

Our existing work is modified to a system with a database which is helpful for attendance marking using face recognition techniques.

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